

Maxwell Alvord — Resume

Full Stack Developer · AI Retrieval & RAG

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Summary

Full-stack developer with 2.5 years across enterprise POS systems (D365 Commerce, RMH, SQL Server) and product work, now building and evaluating AI retrieval (RAG) systems. Strongest on back-end and data work, and comfortable owning a product from database to App Store.

Experience

Independent Full Stack Developer — Self-employed · Portland, OR · Jan 2026 – Present

- Built a reusable, config-driven [small-business site template](#) in Alpine.js with no build step, including live open-hours logic, a filterable menu with lightbox, and a validated contact form, backed by security headers and automated browser tests.
- Shipped two products end to end: Ascoé, an iOS app now live on the App Store, and Money Manager, a public personal finance web app.
- Built a RAG retrieval benchmark over real POS documentation and evaluated four retrieval setups, including Antfly, a Portland retrieval-backend startup's engine.

Systems Engineer — New West Technologies · Portland, OR (Hybrid) · Aug 2023 – Jan 2026

- Set up and smoke-tested D365 Commerce Scale Units and integrated them with the company's payment processing software.
- Managed client SQL Server databases in SSMS, including schema changes, backups, and migrations.
- Built custom modules for the in-house Modern Suite platform and maintained RMH POS systems.
- Led documentation for payment connectors and the D365 lab environment as the company moved into enterprise accounts.

Full Stack Developer — Opine · Remote (Contract) · Oct 2022 – Mar 2023

- Implemented updates and bug fixes in a production React codebase.
- Reviewed peer merge requests before senior review and production deployment.
- Managed Cloud Firestore test data for QA on the staging environment.

Projects

RAG Retrieval Benchmark — Python, sentence-transformers (MiniLM, BGE), BM25, NumPy, Ollama, Antfly, WSL2

- Built a retrieval benchmark over RMH POS documentation using a 19-question, fingerprint-verified golden set scored by MRR, with the LLM removed from the loop to isolate retrieval quality.
- Raised MRR from 0.831 (MiniLM baseline) to 0.860 by upgrading to BGE-base embeddings, and ran a chunking sweep showing one chunk per file wins because each doc is already single-topic.
- Diagnosed a hybrid BM25 + RRF regression (0.737) to BM25 length bias and equal-weight fusion.
- Deployed Antfly locally after tracing embedding failures on an 8 GB laptop to memory pressure; its semantic search scored 0.896 on the same golden set.
- Documented methods, results, and limitations, including an unresolved reranker timeout, in a public GitHub README.

Ascoé — Next.js, Capacitor, Clerk, Neon Postgres, Firebase Cloud Messaging, Vercel · [ascoe.space](#)

- Co-founded and built a cross-gender Q&A app where men and women anonymously ask questions and curate honest answers; live on the App Store.
- Took the app through App Store review, resolving two rejections by redesigning the verification flow and adding a reviewer demo account.
- Implemented push notifications with FCM and APNs, including a daily streak-reminder job and an admin broadcast endpoint.

Money Manager — Next.js, TypeScript, Drizzle ORM, Neon Postgres, Clerk, Vercel · [MoneyManager.live](https://moneymanager.live)

- Built and publicly launched a personal budgeting app with live transaction tracking and spending insights.
- Fixed a data-integrity gap by moving from Neon's HTTP driver to its WebSocket driver so multi-step transactions write atomically.

Skills

Languages: TypeScript, JavaScript, Python, C#, Go, SQL, HTML/CSS

Frameworks: React, Next.js, Node.js, Express, ASP.NET MVC, Astro, Alpine.js, Capacitor, Tailwind CSS

AI & Retrieval: RAG pipelines, embeddings (sentence-transformers), vector similarity search, retrieval evaluation (MRR), Ollama, tool-calling agents, Antfly

Data & Cloud: SQL Server, Postgres (Neon), MySQL, MongoDB, Firebase/Firestore, Drizzle ORM, Vercel, Cloudflare Pages

Enterprise & Tools: D365 Commerce (CSU), RMH POS, Git, Docker, Linux, Proxmox, Jira, Postman, Jest, Selenium

Education

Epicodus — Certificate in Web & Mobile Development

27-week full-time program with 1,000+ hours of classroom and internship work, including 75+ projects in C#, JavaScript, HTML, and CSS built largely through pair programming.